

Chapter 6

Hypothesis Testing: One Sample

Chapter 5 asked: given what we observed, what range of values for μ is plausible? That is estimation. This chapter asks a different question: someone has made a specific claim about μ . Is our sample data consistent with that claim, or does it provide evidence against it?

This is **hypothesis testing**: a formal procedure for deciding whether sample data give sufficient evidence to reject a stated claim about a population parameter.

The machinery is entirely familiar. We will use the CLT, z -scores, and the z -table — the same tools as Chapters 4 and 5. What changes is the question we are answering and the framework we use to answer it.

6.1 The Logic of Hypothesis Testing

6.1.1 A Motivating Example

A coffee machine is supposed to dispense 8.0 oz per cup ($\mu = 8.0$). An inspector suspects the machine is underfilling. She samples 36 cups and finds $\bar{x} = 7.82$ oz, with known $\sigma = 0.5$ oz.

Is $\bar{x} = 7.82$ far enough below 8.0 to be convincing evidence of underfilling, or could this just be normal sampling variation?

From Chapter 4: if μ really is 8.0, then $\bar{x} \sim N(8.0, 0.5/\sqrt{36}) = N(8.0, 0.0833)$. The z -score for the observed mean is:

$$z = \frac{7.82 - 8.0}{0.0833} = \frac{-0.18}{0.0833} \approx -2.16.$$

If the machine really dispenses 8.0 oz on average, a sample mean this low (or lower) would occur with probability $P(Z \leq -2.16) \approx 0.015$. That is only 1.5%.

This small probability is our evidence. It says: *if the claim $\mu = 8.0$ is true, what we observed would be very unlikely*. The decision is whether that is unlikely enough to reject the claim.

6.1.2 The Framework

Definition: Null Hypothesis H_0 . The **null hypothesis** H_0 is the default claim about the population parameter. It is the statement we are testing — the status quo or the manufacturer’s claim. It always contains an equality: $\mu = \mu_0$ or $p = p_0$.

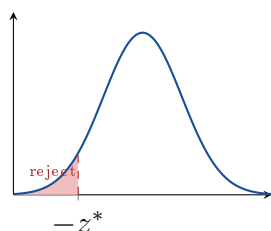
Definition: Alternative Hypothesis H_a . The **alternative hypothesis** H_a (also written H_1) is the claim we are trying to find evidence *for*. It is the research hypothesis, the suspicion, the direction of interest. It contains an inequality: $<$, $>$, or $\neq$.

There are three types of alternative hypotheses, leading to three types of tests:

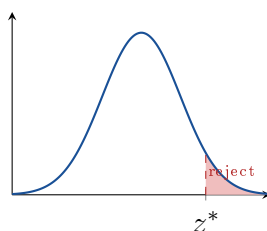
Alternative	Type	Suspicion
$H_a : \mu < \mu_0$	Left-tailed	Parameter is too low
$H_a : \mu > \mu_0$	Right-tailed	Parameter is too high
$H_a : \mu \neq \mu_0$	Two-tailed	Parameter has changed (direction unknown)

Visual Insight. The three test types correspond to which tail(s) of the normal curve count as “unusual.”

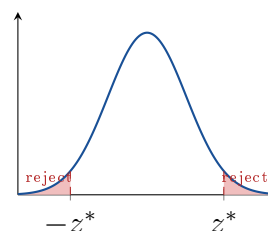
Left-tailed: $H_a : \mu < \mu_0$



Right-tailed: $H_a : \mu > \mu_0$



Two-tailed: $H_a : \mu \neq \mu_0$



The red shaded regions are the **rejection regions**: where the test statistic must fall to reject H_0 . The size of the rejection region is exactly α , the significance level.

Exercises 6.1

- 6.1.1. For each scenario, state H_0 and H_a , and identify the test type (left, right, or two-tailed). (a) A manufacturer claims average battery life is 500 hours. A consumer group suspects it is less. (b) A school district’s average reading score has historically been 72. A new curriculum is introduced; the district wants to know if the average has changed. (c) A drug company claims its pill reduces LDL

cholesterol by 30 points on average. A skeptic believes the reduction is smaller.

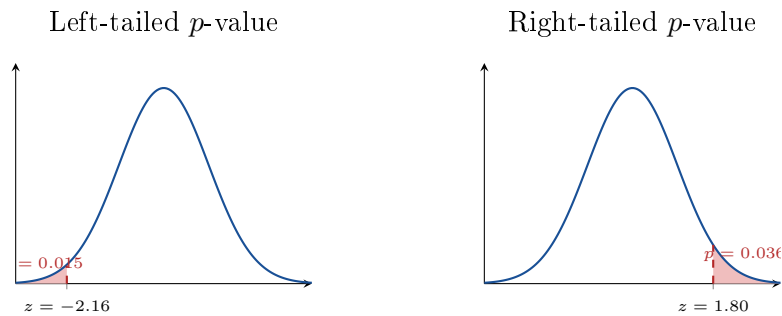
6.1.2. Why must the null hypothesis contain an equality ($=$) rather than an inequality ($<$, $>$, $\neq$)?

6.2 The p -Value

Definition: p -Value. The p -value is the probability of obtaining a test statistic as extreme as (or more extreme than) the one observed, *assuming H_0 is true*.

A small p -value means: if H_0 were true, what we observed would be very unlikely. This is evidence against H_0 .

Visual Insight. The p -value is an area under the standard normal curve. The test statistic z is a point on that curve. The p -value is the area in the tail(s) beyond z .



6.2.1 Computing the p -Value

Test type	p -value
Left-tailed ($H_a : \mu < \mu_0$)	$P(Z \leq z_{\text{obs}})$
Right-tailed ($H_a : \mu > \mu_0$)	$P(Z \geq z_{\text{obs}}) = 1 - P(Z \leq z_{\text{obs}})$
Two-tailed ($H_a : \mu \neq \mu_0$)	$2 \cdot P(Z \geq z_{\text{obs}}) = 2(1 - P(Z \leq z_{\text{obs}}))$

6.2.2 Significance Level and the Decision Rule

Definition: Significance Level α . The **significance level** α is a threshold chosen *before* the test. Common choices: $\alpha = 0.05$ (5%) or $\alpha = 0.01$ (1%).

Decision rule:

- If $p\text{-value} \leq \alpha$: **Reject** H_0 . The data provide sufficient evidence for H_a .
- If $p\text{-value} > \alpha$: **Fail to reject** H_0 . The data do not provide sufficient evidence for H_a .

Common Pitfall. “Fail to reject H_0 ” is not the same as “accept H_0 .” We never prove the null hypothesis true; we only fail to find evidence against it. The absence of evidence is not evidence of absence. A trial that finds insufficient evidence to convict does not prove innocence — it only means we could not prove guilt beyond the standard.

Exercises 6.2

- 6.2.1. A test produces $z = -1.88$. Compute the p -value for: (a) a left-tailed test; (b) a right-tailed test; (c) a two-tailed test. For each, draw the normal curve and shade the p -value region.
- 6.2.2. A p -value is 0.032. State the decision at $\alpha = 0.05$. State the decision at $\alpha = 0.01$. Explain the difference.
- 6.2.3. A researcher reports a p -value of 0.08 and concludes “We have proven H_0 is true.” Identify and correct the error.

6.3 One-Sample z -Test for a Mean

Key Formula. One-sample z -test for μ (when σ is known, $n \geq 30$):

Hypotheses: $H_0 : \mu = \mu_0$ vs. $H_a : \mu \neq \mu_0$ (or $<$ or $>$).

Test statistic:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}.$$

p -value: from the z -table, based on test type.

Decision: reject H_0 if $p\text{-value} \leq \alpha$.

The test statistic z answers: how many standard errors is $\bar{x}$ from the claimed value μ_0 ? The further $\bar{x}$ is from μ_0 , the more evidence against the null.

6.3.1 The Four-Step Process

Every hypothesis test follows the same four steps:

1. **State:** Write H_0 and H_a . Choose α . Identify test type. Check conditions ($n \geq 30$, σ known).
2. **Test statistic:** Compute $z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$.
3. **p-value:** Compute and sketch.
4. **Conclude:** Compare p to α . State the conclusion in context.

Example 6.3.1. (*Left-Tailed Test: Coffee Machine*) Returning to our motivating example. Claim: $\mu = 8.0$ oz. Inspector suspects underfilling. Sample: $n = 36$, $\bar{x} = 7.82$ oz, $\sigma = 0.5$ oz. Use $\alpha = 0.05$.

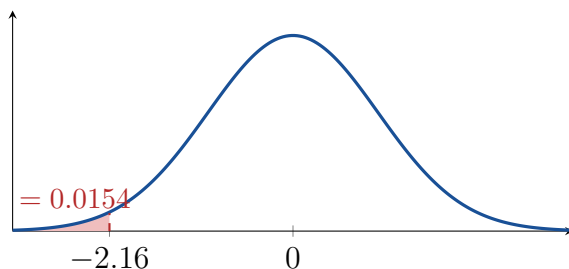
Step 1. $H_0 : \mu = 8.0$ oz. $H_a : \mu < 8.0$ oz (left-tailed). $n = 36 \geq 30$. ✓

Step 2.

$$z = \frac{7.82 - 8.0}{0.5/\sqrt{36}} = \frac{-0.18}{0.0833} \approx -2.16.$$

Step 3. Left-tailed: $p = P(Z \leq -2.16) = 0.0154$.

Left-tailed test: p -value = 0.0154



Step 4. $p = 0.0154 < \alpha = 0.05$. **Reject H_0 .**

Conclusion: There is sufficient evidence at the 5% significance level to conclude that the machine is underfilling cups (the true mean is less than 8.0 oz).

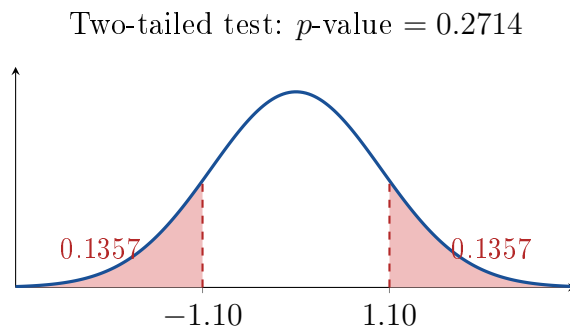
Example 6.3.2. (*Two-Tailed Test: Exam Scores*) A school district claims the average score on a statewide exam is $\mu_0 = 500$. A researcher suspects this has changed. She samples 64 students: $\bar{x} = 511$, $\sigma = 80$. Test at $\alpha = 0.05$.

Step 1. $H_0 : \mu = 500$. $H_a : \mu \neq 500$ (two-tailed). $n = 64 \geq 30$. ✓

Step 2.

$$z = \frac{511 - 500}{80/\sqrt{64}} = \frac{11}{10} = 1.10.$$

Step 3. Two-tailed: $p = 2(1 - P(Z \leq 1.10)) = 2(1 - 0.8643) = 2(0.1357) = 0.2714$.



Step 4. $p = 0.2714 > \alpha = 0.05$. **Fail to reject H_0 .**

Conclusion: There is insufficient evidence to conclude that the mean score has changed from 500. A sample mean of 511 is not unusual enough to be convincing.

Exercises 6.3

6.3.1. A bottle-filling machine should deliver $\mu_0 = 16.0$ oz. A technician samples 50 bottles: $\bar{x} = 16.08$ oz, $\sigma = 0.2$ oz. Test $H_0 : \mu = 16.0$ vs. $H_a : \mu > 16.0$ at $\alpha = 0.05$. (a) Compute z . (b) Find the p -value. (c) State the conclusion.

6.3.2. Using the same data, test at $\alpha = 0.01$. Does the conclusion change?

6.3.3. A researcher tests $H_0 : \mu = 100$ and gets $z = -1.75$. (a) Find the two-tailed p -value. (b) At $\alpha = 0.05$, what is the conclusion? (c) If instead $H_a : \mu < 100$, does the conclusion change? Why?

6.4 One-Sample z -Test for a Proportion

Key Formula. One-sample z -test for p :

Hypotheses: $H_0 : p = p_0$ vs. $H_a : p \neq p_0$ (or $<$ or $>$).

Test statistic:

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}.$$

Conditions: $np_0 \geq 10$ and $n(1 - p_0) \geq 10$.

Notice: the standard error here uses p_0 (the claimed value), not $\hat{p}$. This is the opposite of the confidence interval, which used $\hat{p}$ because there was no claimed value. In hypothesis testing, we assume H_0 is true to build the null distribution, so we use p_0 .

Example 6.4.1. (*One-Sample z -Test for a Proportion*) A seed company claims 85% of its seeds germinate. A gardener plants 200 seeds and only 158 germinate. Is there evidence at $\alpha = 0.05$ that the true germination rate is less than 85%?

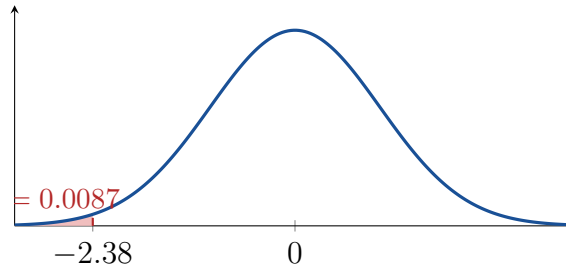
Step 1. $H_0 : p = 0.85$. $H_a : p < 0.85$ (left-tailed). $\hat{p} = 158/200 = 0.79$. Check: $200(0.85) = 170 \geq 10$, $200(0.15) = 30 \geq 10$. ✓

Step 2.

$$z = \frac{0.79 - 0.85}{\sqrt{0.85 \cdot 0.15/200}} = \frac{-0.06}{\sqrt{0.000638}} = \frac{-0.06}{0.02525} \approx -2.38.$$

Step 3. Left-tailed: $p = P(Z \leq -2.38) = 0.0087$.

Left-tailed test for proportion: p -value = 0.0087



Step 4. $p = 0.0087 < \alpha = 0.05$. **Reject H_0 .**

Conclusion: There is sufficient evidence at the 5% level to conclude that the true germination rate is less than 85%.

Common Pitfall. In the proportion test, the standard error uses the **null** value p_0 :

$$SE = \sqrt{\frac{p_0(1-p_0)}{n}}, \quad \text{NOT} \quad \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

We use p_0 because hypothesis testing asks “how surprising is this result *if* H_0 were true?” We build the distribution assuming the null is true.

Exercises 6.4

6.4.1. A university claims 60% of its graduates find employment in their field within 6 months. A sample of 150 graduates finds 81 employed in their field. Test $H_0 : p = 0.60$ vs. $H_a : p < 0.60$ at $\alpha = 0.05$. Follow all four steps.

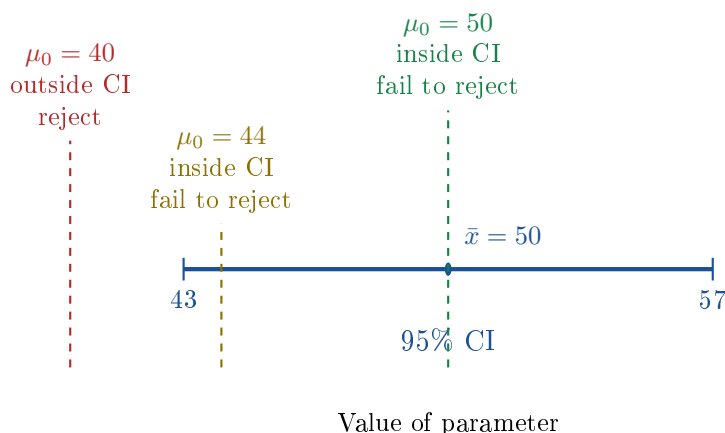
6.4.2. A coin is flipped 100 times and lands heads 58 times. Test whether the coin is fair ($H_0 : p = 0.5$, $H_a : p \neq 0.5$) at $\alpha = 0.05$.

6.4.3. In Exercise 2, what is the minimum number of heads in 100 flips that would lead to rejecting H_0 at $\alpha = 0.05$ (two-tailed)?

6.5 The Connection Between Hypothesis Tests and Confidence Intervals

Hypothesis tests and confidence intervals are two sides of the same coin. A two-tailed hypothesis test at significance level α rejects $H_0 : \mu = \mu_0$ if and only if μ_0 falls *outside* the corresponding $(1 - \alpha)$ confidence interval.

Visual Insight. The figure below shows the same data analyzed both ways. The 95% CI is shown as a horizontal bar. Three possible claimed values (μ_0) are tested:



Any μ_0 inside the 95% CI would *not* be rejected at $\alpha = 0.05$ (two-tailed). Any μ_0 outside would be rejected. The CI is the set of all values of μ_0 for which H_0 would fail to be rejected.

This connection is worth holding onto. When you compute a 95% CI and it does not contain some claimed value, you have implicitly already rejected that claim at the 5% significance level.

Chapter 6 Review

1. The mean resting heart rate of adults is claimed to be 72 bpm. A cardiologist suspects patients in a high-stress occupation have a higher rate. She samples 45 patients: $\bar{x} = 75.8$ bpm, $\sigma = 12$ bpm. (a) State H_0 and H_a . Identify the test type. (b) Compute the test statistic z . (c) Find and sketch the p -value. (d) State the conclusion at $\alpha = 0.05$ and at $\alpha = 0.01$.
2. A consumer watchdog group tests whether a fast-food chain's advertised calorie count (540 calories) is accurate. They sample 36 items: $\bar{x} = 562$ cal, $\sigma = 60$ cal. (a) Test at $\alpha = 0.05$, two-tailed. (b) Compute a 95% CI for μ . Does the advertised value 540 lie inside? (c) Confirm that your test result and your CI are consistent.

3. A poll claims 55% of voters in a district support a ballot measure. A rival campaign surveys 200 voters and finds 100 in support. (a) Test $H_0 : p = 0.55$ vs. $H_a : p < 0.55$ at $\alpha = 0.05$. (b) Test $H_0 : p = 0.55$ vs. $H_a : p \neq 0.55$ at $\alpha = 0.05$. (c) The p -value for (b) is exactly twice the p -value for (a). Explain why, using the picture.
4. *Big picture:* A study finds $p = 0.049$. A journalist writes: “Scientists have proven their hypothesis true at the 95% confidence level.” Write a one-paragraph correction of all the statistical errors in this sentence.